

Variables & Data Types

1. Declare two integer variables, a and b. Swap their values using a temporary third variable and print the values before and after the swap.
2. Declare two integer variables, a and b. Swap their values using only arithmetic operations (+, -) and print the values before and after the swap.
3. Ask the user for the length and width of a rectangle. Compute and print both its area and perimeter.
4. Take the radius of a circle as input from the user and calculate its area using the formula $\text{Area} = \pi * r^2$.
5. Write a program that converts a given number of days into years, weeks, and remaining days. (Assume 1 year = 365 days).
6. Unit Converter:
 - Convert a given weight in kilograms to grams.
 - Convert a given distance in metres to centimetres.
 - Convert a given amount in US dollars to BDT (use a fixed conversion rate, e.g., 1 USD = 117 BDT).
6. Take the principal amount, annual interest rate, and time (in years) as input. Calculate and print the simple interest using the formula $SI = (P \times R \times T) / 100$.
7. Convert a temperature from Celsius to Fahrenheit. The formula is $F = (C \times 9 / 5) + 32$.
8. Convert a temperature from Fahrenheit to Celsius. The formula is $C = (F - 32) \times 5 / 9$.
9. Take marks for three subjects as input and calculate the average mark.
10. Declare a char variable and print both the character and its corresponding ASCII value.
11. Ask the user for their first name and last name. Concatenate them to form a full name and print it.
12. Take the length, width, and height of a cuboid as input and calculate its volume.
13. Declare variables of type int, double, char, and boolean. Assign them values and print them out with a label for each.

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